

MICROOPERATIONS

- Computer system microoperations are of four types:

- Register transfer microoperations
- Arithmetic microoperations
- Logic microoperations
- Shift microoperations

Register contents get changed with these operations

We have covered this portion. The content of the register do not change during the transfer.



ARITHMETIC MICROOPERATIONS

- The basic arithmetic microoperations are
 - Addition
 - Subtraction
 - Increment
 - Decrement
 - Shift
- The additional arithmetic microoperations are
 - Add with carry
 - Subtract with borrow
 - Transfer/Load
 - etc. ...

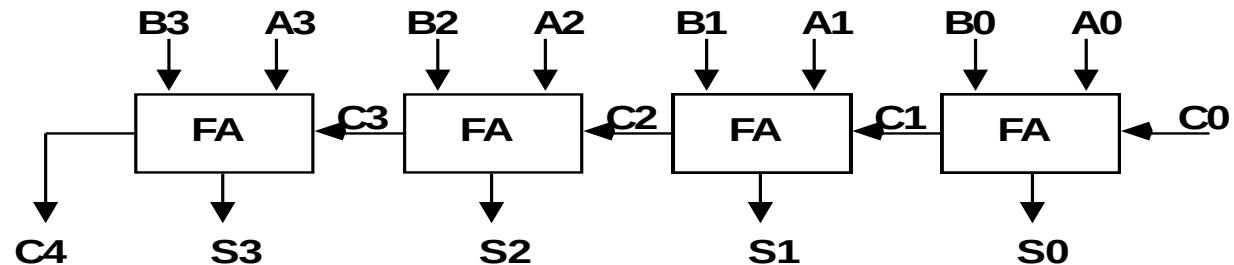
Summary of Typical Arithmetic Micro-Operations

$R3 \leftarrow R1 + R2$	Contents of R1 plus R2 transferred to R3
$R3 \leftarrow R1 - R2$	Contents of R1 minus R2 transferred to R3
$R2 \leftarrow R2'$	Complement the contents of R2 (1's Complement)
$R2 \leftarrow R2' + 1$	2's complement the contents of R2 (Negate)
$R3 \leftarrow R1 + R2' + 1$	R1 plus 2's Complement of R2 (Subtraction)
$R1 \leftarrow R1 + 1$	Increment
$R1 \leftarrow R1 - 1$	Decrement

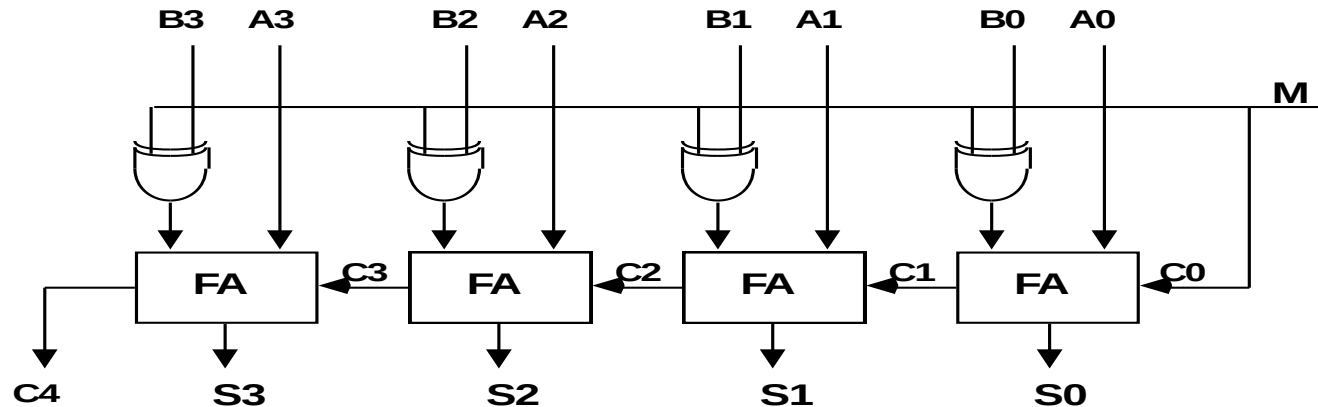


BINARY ADDER / SUBTRACTOR / INCREMENTER

Binary Adder



Binary Adder-Subtractor



Binary Incrementer

